

Conditional Statements(Decision making) - Zero to Hero Questions List

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Assuming you have already learned **variables, data types, input/output, arithmetic operators, relational operators, logical operators, assignment operators, and conditional statements (if, if-else, else-if, nested if, switch)**, here are practice problems from beginner to advanced.

1 Beginner Level

1. Check whether a number is positive or negative.
 - **Hint:** Use **if-else**.
2. Check whether a number is even or odd.
 - **Hint:** Use **% 2**.
3. Find the greater of two numbers.
 - **Hint:** Compare using **>**.
4. Find the smaller of two numbers.
 - **Hint:** Compare using **<**.
5. Check whether a person is eligible to vote.
 - **Hint:** Age **>= 18**.
6. Check whether a number is divisible by 5.
 - **Hint:** Use **%**.
7. Check whether a number is divisible by both 2 and 3.
 - **Hint:** Use **&&**.
8. Check whether a character is a vowel.
 - **Hint:** Compare with **a, e, i, o, u**.
9. Check whether a character is an uppercase letter.
 - **Hint:** ASCII range **'A' to 'Z'**.
10. Check whether a character is a lowercase letter.
 - **Hint:** ASCII range **'a' to 'z'**.

2 Easy Level

11. Find the largest of three numbers.
 - **Hint:** Use `if-else-if`.
 12. Find the smallest of three numbers.
 - **Hint:** Multiple comparisons.
 13. Check whether a year is a leap year.
 - **Hint:** Divisible by 4, 100, and 400.
 14. Check whether a number lies between 10 and 100.
 - **Hint:** Use `&&`.
 15. Check whether a number is a multiple of 7.
 - **Hint:** `% 7 == 0`.
 16. Check whether a character is an alphabet.
 - **Hint:** Check uppercase and lowercase ranges.
 17. Check whether a character is a digit.
 - **Hint:** Range `'0'` to `'9'`.
 18. Check whether a character is an alphabet, digit, or special character.
 - **Hint:** Nested conditions.
 19. Calculate profit or loss.
 - **Hint:** Compare cost price and selling price.
 20. Determine whether a student passed or failed.
 - **Hint:** Marks `>` passing marks.
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3 Medium Level

21. Find the grade based on marks.
 - **Hint:** Use `else-if` ladder.
22. Find the largest among four numbers.
 - **Hint:** Multiple conditions.
23. Check whether three sides can form a triangle.
 - **Hint:** Triangle inequality theorem.

24. Determine triangle type.
 - **Hint:** Equilateral, Isosceles, Scalene.
 25. Check whether a number is a 2-digit, 3-digit, or 4-digit number.
 - **Hint:** Use ranges.
 26. Check whether a number is divisible by either 3 or 5.
 - **Hint:** Use `||`.
 27. Determine employee bonus based on years of service.
 - **Hint:** Multiple ranges.
 28. Calculate electricity bill based on units consumed.
 - **Hint:** Slab system.
 29. Calculate income tax based on salary slabs.
 - **Hint:** Use nested conditions.
 30. Check whether a character is a vowel or consonant.
 - **Hint:** First verify alphabet.
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4 Switch Statement Practice

31. Build a simple calculator using `switch`.
 - **Hint:** Operator as input.
32. Display day of the week from day number.
 - **Hint:** Cases 1–7.
33. Display month name from month number.
 - **Hint:** Cases 1–12.
34. Find number of days in a month.
 - **Hint:** Use `switch`.
35. Menu-driven arithmetic operations.
 - **Hint:** User chooses operation.
36. Convert lowercase vowel to uppercase.
 - **Hint:** Switch on character.
37. Display department based on code.
 - **Hint:** Character codes.
38. Display grade description using `switch`.

- **Hint:** Cases A, B, C, D.
39. Simple ATM menu.
- **Hint:** Menu options.
40. Restaurant menu billing system.
- **Hint:** Item codes.
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5 Advanced Level

41. Find the largest among five numbers.
- **Hint:** Nested `if`.
42. Check whether a point lies in Quadrant I, II, III, or IV.
- **Hint:** Check signs of `x` and `y`.
43. Determine roots of a quadratic equation.
- **Hint:** Check discriminant.
44. Check whether three angles form a valid triangle.
- **Hint:** `Sum = 180`.
45. Check whether a number belongs to a specific range category.
- **Hint:** Multiple intervals.
46. Determine scholarship eligibility.
- **Hint:** Combine marks and income.
47. Check whether a year is a century leap year.
- **Hint:** Special leap year rules.
48. Determine ticket fare based on age category.
- **Hint:** Child, Adult, Senior Citizen.
49. Check whether a number is divisible by 2, 3, 5, and 7.
- **Hint:** Multiple conditions.
50. Build a complete student result system.
- **Hint:** Total, percentage, pass/fail, grade.
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6 Challenge Problems

51. Find the second largest among three numbers.
 - **Hint:** Largest is not enough.
52. Determine zodiac sign from birth month and day.
 - **Hint:** Range checking.
53. Validate a date (DD/MM/YYYY).
 - **Hint:** Month-wise day limits.
54. Determine season from month number.
 - **Hint:** Group months.
55. Check whether a point lies on the X-axis, Y-axis, Origin, or Quadrant.
 - **Hint:** Special cases first.
56. Determine shipping charges based on weight and distance.
 - **Hint:** Nested slabs.
57. Calculate discount based on purchase amount.
 - **Hint:** Multiple discount slabs.
58. Determine insurance premium category.
 - **Hint:** Age + vehicle type.
59. Check admission eligibility based on multiple subjects.
 - **Hint:** Combined conditions.
60. Create a menu-driven mini application using nested switches.
 - **Hint:** Main menu → Submenu.